

PhySense-AdjointFlow: Direct Endpoint Trajectory Optimization via Differentiable ODE Unrolling for Sparse Physical Sensing

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Abstract

Physical field reconstruction from sparse sensor observations is a core challenge across fluid dynamics, oceanography, and aerospace engineering. Recent flow-based sensing co-optimization paradigms, such as PhySense, jointly learn a field reconstructor and optimize sensor placements using a single-step instantaneous flow-matching surrogate loss $\mathcal{L}_{\text{flow}}(p)$. However, field generation at inference time relies on numerical integration along an N -step Ordinary Differential Equation (ODE) trajectory ($X_0 \rightarrow \bar{X}_1$), creating a critical objective mismatch: single-step velocity matching ignores the compounding, non-linear discretization drift that accumulates during multi-step integration. In this work, we propose PhySense-AdjointFlow, a retraining-free sensor placement optimization paradigm that backpropagates directly through an N -step unrolled numerical ODE solver to minimize the test-time endpoint field reconstruction error $\mathcal{L}_{\text{endpoint}}(p) = \mathbb{E}[\|\bar{X}_1(p) - X_1\|_2^2]$. Because the velocity network v_θ remains frozen during placement updates, backpropagating with respect to the low-dimensional sensor coordinates $p \in \mathbb{R}^{m \times d}$ retains a minimal ~ 1 -step activation memory footprint (≈ 14.5 GB) via gradient checkpointing. We further introduce a hybrid objective with linear warm-up decay that supplies smooth global spatial gradients during early projected gradient descent iterations. Under a strictly matched-initialization, matched-protocol comparison on 2D Turbulent Channel Flow and global Sea Surface Temperature benchmarks, PhySense-AdjointFlow consistently outperforms the single-step flow surrogate, reducing relative L_2 reconstruction error by up to 5.4% under low sensor coverage ($m = 30$), with paired improvements that are statistically significant across five random initializations ($p < 0.01$ at every budget) and robust across a $1 \times -5 \times$ range of test-time integration step counts. A runtime study confirms that placement search runs at constant ≈ 14.5 GB memory regardless of unroll depth, making exact trajectory optimization practical.

1 Introduction

Reconstructing continuous physical fields—such as turbulent fluid flows, global ocean sea surface temperatures, and aerodynamic surface pressure distributions—from sparse sensor measurements is a foundational challenge across computational physics, fluid dynamics, oceanography, and aerospace engineering (Ongie et al., 2020; Gupta et al., 2026). Because physical experiments and high-density sensor deployments are constrained by severe hardware and financial budgets, selecting optimal spatial locations for a limited number of sensors is critical to maximizing observational information gain and ensuring high-fidelity field reconstruction (Huan & Marzouk, 2011; Ramos et al., 2026).

Recent advances in generative modeling, particularly continuous-time frameworks such as Continuous Normalizing Flows (Chen et al., 2018) and Flow Matching (Lipman et al., 2022; Albergo & Vanden-Eijnden, 2022), have established a compelling paradigm for solving non-linear inverse problems in physical systems. Building on these flow-based architectures,

PhySense (Ma et al., 2025) recently introduced a co-optimization paradigm that couples a cross-attention flow reconstructor with geometry-constrained projected gradient descent (PGD) over sensor coordinates. To optimize sensor positions, PhySense minimizes a single-step instantaneous flow-matching loss $\mathcal{L}_{\text{flow}}(p)$. However, physical field generation at test time requires numerically integrating the learned velocity field along an N -step ODE trajectory ($X_0 \rightarrow \bar{X}_1$). This divergence introduces an objective mismatch (Lambert et al., 2020): optimizing an instantaneous single-step velocity field does not account for the compounding, non-linear trajectory errors that accumulate during test-time multi-step integration, where late-stage velocity errors exert disproportionate influence on the final reconstructed field.

To resolve this mismatch, we introduce PhySense-AdjointFlow, a framework that directly optimizes sparse sensor placements against the exact test-time endpoint trajectory reconstruction error $\mathcal{L}_{\text{endpoint}}(p) = \mathbb{E}_{X_0, X_1}[\|\bar{X}_1(p) - X_1\|_2^2]$. Our key insight is that during placement optimization the parameters of the learned reconstructor v_θ remain frozen. Consequently, backpropagating through an N -step unrolled ODE solver solely with respect to the low-dimensional sensor coordinates $p \in \mathbb{R}^{m \times d}$ requires no velocity-network weight gradients $\nabla_\theta v_\theta$. We stress that this is a discrete adjoint—i.e., exact reverse-mode backpropagation-through-time (BPTT) over the unrolled explicit-Euler steps with per-step activation checkpointing—rather than a continuous adjoint sensitivity ODE (e.g., Pontryagin’s principle in Neural ODEs); we retain the name AdjointFlow for its discrete-adjoint trajectory-differentiation character. Combined with per-step gradient checkpointing, differentiable trajectory unrolling incurs a near-single-step memory footprint while aligning placement gradients with the exact multi-step generation process used at inference. Notably, the original framework avoided this objective precisely because of assumed computational overhead; we show this overhead effectively vanishes when the reconstructor is frozen.

To stabilize optimization and provide smooth spatial guidance when sensors are initialized far from salient flow features, PhySense-AdjointFlow employs a hybrid objective $\mathcal{L}_{\text{hybrid}}(p) = \mathcal{L}_{\text{endpoint}}(p) + \lambda_t \cdot \mathcal{L}_{\text{flow}}(p)$, where the single-step warm-up weight λ_t is linearly decayed across optimization epochs. Placements are iteratively updated via PGD to respect physical boundary constraints.

We validate PhySense-AdjointFlow under a strict apples-to-apples protocol: both the flow-surrogate control and our method start from the same random initialization, the same frozen official reconstructor, and the same official PGD walk protocol—the only difference is the placement objective. On 2D turbulent flow under low coverage ($m = 30$), PhySense-AdjointFlow attains a test relative L_2 error of 0.2080, improving on random sampling (0.5296, -60.7%) and, crucially, on the single-step flow control (0.2197, a 5.4% relative error reduction), with consistent gains across $m \in \{50, 100, 200\}$. Our contributions are:

- A retraining-free trajectory-aligned placement objective. We formulate sensor placement optimization as direct minimization of the N -step unrolled endpoint reconstruction error, eliminating the single-step surrogate gap without any geometric spread priors or heuristic forces (Section 3).
- A memory-efficient realization. By exploiting the frozen reconstructor and per-step gradient checkpointing, deep ODE unrolling stays at ~ 1 -step activation memory (≈ 14.5 GB), making the “prohibitively expensive” exact objective tractable.
- Faithful, comprehensive evaluation and diagnostics. We report matched-protocol gains on turbulent flow and sea temperature with multi-seed paired significance testing ($p < 0.01$ at every budget) and a wall-clock/memory profile confirming constant-memory search, and dissect why the method works: a two-pathway gradient analysis identifying observation-value sampling as the dominant driver, and a step-count robustness study showing placements transfer across a $1 \times -5 \times$ range of inference steps (Section 4).

2 Related Work

Classical and statistical sensor placement. Optimal sensor placement has historically been grounded in optimal experimental design (Huan & Marzouk, 2011) and statistical estimation

theory. A prominent paradigm is Sparse Sensor Placement for Optimal Reconstruction (SSPOR) (Manohar et al., 2017), which identifies sensor locations via QR pivoting over low-rank modal bases from Proper Orthogonal Decomposition or Singular Value Decomposition. Related sparse-representation methods leverage ℓ_1 minimization over spatial dictionaries for robust flow estimation from limited measurements (Callahan et al., 2018). In geophysical regimes, data-assimilation techniques recover dense state fields from sparse observations (Gupta et al., 2026). These frameworks, however, fundamentally rely on linear-Gaussian assumptions, static covariance matrices, or linear modal truncations, and thus struggle with non-stationary multi-scale turbulence and complex boundary geometries without hand-crafted heuristics.

Deep generative and neural field reconstructors. To overcome the expressiveness limits of linear methods, deep architectures have been widely adopted for physics reconstruction. Deterministic operator networks such as Senseiver (Santos et al., 2023) and FLRONet (Dang et al., 2024) encode scattered observations via cross-attention to reconstruct dense fields over arbitrary geometries, and recent work couples transport-aware graph networks with reinforcement-learned placement policies (Li et al., 2026). Generative approaches include Diff-SPORT (Vishwasrao et al., 2025), which pairs score-based diffusion with maximum-a-posteriori inference for urban turbulence, and bias-constrained schedules that mitigate exposure bias in unrolled PDE diffusion emulators (Cleï et al., 2026). PhySense (Ma et al., 2025) established a co-optimization framework in which a cross-attention flow reconstructor supplies reconstruction feedback for gradient-based placement. Yet existing reconstructors either decouple placement into a static pre-processing step or optimize it with single-step local surrogates; as formalized in the objective-mismatch literature (Lambert et al., 2020) and recent flow fine-tuning studies (Li et al., 2025), minimizing single-step vector-field metrics fails to capture multi-step trajectory error accumulation, yielding sub-optimal placements at test time.

Flow matching and differentiable ODE optimization. Continuous-time generative modeling via Continuous Normalizing Flows (Chen et al., 2018) and Flow Matching (Lipman et al., 2022; Albergo & Vanden-Eijnden, 2022) enables simulation-free training by regressing velocity fields along target probability paths such as optimal-transport interpolants. Standard flow matching nonetheless exhibits a train-inference gap: models are trained on instantaneous velocity losses but evaluated via multi-step ODE integration. To close this gap in generative modeling, recent works supervise integrated displacements over trajectory segments (Ahmed et al., 2026), enable direct jumps along the probability-flow ODE (Kim et al., 2024), regularize neural-ODE integration for stability (Finlay et al., 2020), and fine-tune integrated reconstructions by maximum likelihood (Li et al., 2025). In inverse problems and image editing, differentiable ODE unrolling backpropagates through solver steps to optimize input conditioning (Elarabawy et al., 2022). Building on these foundations, PhySense-AdjointFlow applies differentiable trajectory unrolling specifically over low-dimensional sensor coordinates: by freezing the pre-trained reconstructor, it avoids heavy weight-gradient storage while directly eliminating the surrogate objective gap in flow-based physics sensing.

3 Methodology

3.1 Problem Formulation and Single-Step Flow Placement Limits

Let $\mathcal{D} \subset \mathbb{R}^d$ be a bounded spatial domain, and let $X_1 \in \mathbb{R}^{|\mathcal{D}| \times C}$ denote a dense target field with C channels. We place m point sensors at positions $p = \{p_1, \dots, p_m\} \in \mathcal{D}^m$, which gather sparse measurements $X_1(p) \in \mathbb{R}^{m \times C}$. The sensing task is to estimate the full field X_1 from the sparse observations $X_1(p)$ and their coordinates p .

In continuous-time flow matching (Lipman et al., 2022; Ma et al., 2025), reconstruction is modeled as an optimal-transport probability flow mapping a Gaussian source $\pi_0 = \mathcal{N}(0, \mathbf{I})$ to the target distribution π_1 . With the linear interpolant $X_t = tX_1 + (1-t)X_0$ for $t \in [0, 1]$, a conditional velocity field $v_\theta(X_t, t, p, X_1(p))$ is trained by regressing instantaneous

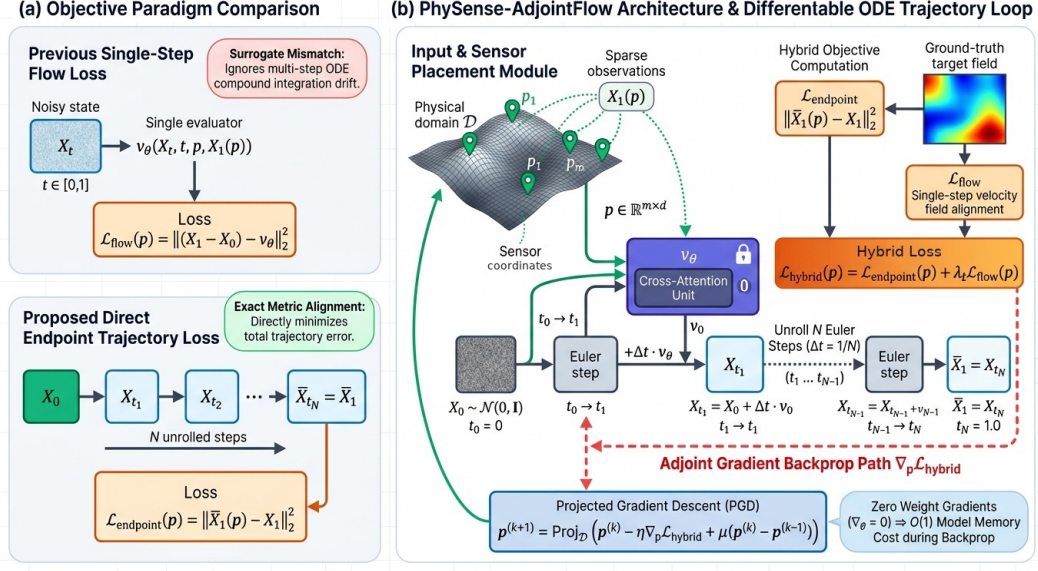


Figure 1: Overview of PhySense-AdjointFlow. (a) Objective comparison: the proposed N -step unrolled endpoint trajectory loss $\mathcal{L}_{\text{endpoint}}(p)$ resolves the compounding integration drift and surrogate mismatch of the single-step flow-matching loss $\mathcal{L}_{\text{flow}}(p)$. (b) Optimization loop: forward N -step Euler integration through a frozen reconstructor v_θ , hybrid loss computation, adjoint trajectory gradient backpropagation directly to the low-dimensional sensor coordinates p , and domain-constrained projected gradient updates. Since $\nabla_\theta v_\theta = \mathbf{0}$, the backward pass costs $\mathcal{O}(1)$ model memory.

displacements:

$$\mathcal{L}_{\text{flow}}(p) = \int_0^1 \mathbb{E}_{X_0, X_1} \left[\|(X_1 - X_0) - v_\theta(X_t, t, p, X_1(p))\|_2^2 \right] dt. \quad (1)$$

Prior flow-based co-optimization such as PhySense (Ma et al., 2025) treats $\mathcal{L}_{\text{flow}}(p)$ as a static surrogate and updates p via PGD. While lightweight, this exhibits a fundamental objective mismatch (Lambert et al., 2020). Evaluating $\mathcal{L}_{\text{flow}}(p)$ requires the velocity at a single sampled time $t \sim \mathcal{U}(0, 1)$ along the ground-truth segment X_t . At test time, however, X_t is unavailable; the field $\bar{X}_1$ is generated by integrating v_θ along a discrete multi-step ODE trajectory from pure noise X_0 . Single-step velocity matching therefore ignores the compounding non-linear discretization errors accumulated across solver steps and assigns uniform weight over time, whereas late-stage steps disproportionately impact the final field.

3.2 Differentiable ODE Trajectory Unrolling and Direct Endpoint Loss

To align placement updates with test-time fidelity, PhySense-AdjointFlow evaluates candidates by fully unrolling the discrete ODE integration during the forward pass. For a configuration $p \in \mathcal{D}^m$ with observations $X_1(p)$, we simulate generation over N explicit Euler steps with $\Delta t = 1/N$ and nodes $t_k = k/N$:

$$X_{t_0} = X_0 \sim \mathcal{N}(0, \mathbf{I}), \quad X_{t_{k+1}} = X_{t_k} + \Delta t \cdot v_\theta(X_{t_k}, t_k, p, X_1(p)). \quad (2)$$

The endpoint field is $\bar{X}_1(p) = X_{t_N}$, and we minimize the direct endpoint reconstruction error:

$$\mathcal{L}_{\text{endpoint}}(p) = \mathbb{E}_{X_0, X_1} \left[\|\bar{X}_1(p) - X_1\|_2^2 \right] = \mathbb{E}_{X_0, X_1} \left[\|X_{t_N}(p) - X_1\|_2^2 \right]. \quad (3)$$

During placement optimization the reconstructor weights θ are frozen ($\nabla_\theta v_\theta = \mathbf{0}$), so backpropagating the scalar loss through the N -step unrolled graph yields spatial gradients $\nabla_p \mathcal{L}_{\text{endpoint}}$ purely with respect to the $m \times d$ coordinates p . Concretely, this is a discrete

Table 1: Summary of physical field reconstruction benchmarks. #Mesh is the number of discretized spatial points per sample.

Benchmark	Geometry	Dim	#Mesh	Dataset Size
2D Turbulent Channel Flow	Regular grid	2D	$128 \times 48 = 6,144$	10,000 frames
Sea Surface Temperature	Land constraints	2D	$360 \times 180 = 64,800$	9,843 train / 730 test

adjoint: exact reverse-mode differentiation (discrete backpropagation-through-time, BPTT) through the unrolled explicit-Euler recurrence in Eq. 2, as opposed to integrating a continuous adjoint sensitivity differential equation. Applying activation gradient checkpointing at each solver step discards the backbone’s intermediate activations during the rollout and recomputes them on demand in the adjoint (backward) pass. Because no weight gradients are accumulated, peak memory remains bounded at ~ 1 -step activation memory (≈ 14.5 GB for $N = 10$ Euler steps on the 128×48 grid), rendering deep unrolling tractable.

3.3 Hybrid Optimization Objective and Warm-Up Schedule

When p is initialized far from salient features, $\mathcal{L}_{\text{endpoint}}(p)$ can exhibit sharp local variations and weak long-range gradients. We therefore construct a hybrid loss combining the global guidance of single-step velocity matching with the exact endpoint criterion:

$$\mathcal{L}_{\text{hybrid}}(p) = \mathcal{L}_{\text{endpoint}}(p) + \lambda_t \cdot \mathcal{L}_{\text{flow}}(p), \quad \lambda_t = \lambda_0 \left(1 - \frac{k}{K-1}\right), \quad (4)$$

with $\lambda_0 = 1.0$ decayed linearly across K PGD epochs ($k \in \{0, \dots, K-1\}$). Early on ($k \approx 0$), $\mathcal{L}_{\text{flow}}(p)$ provides broad long-range attraction to guide scattered sensors toward regions of high flow activity; as $\lambda_t \rightarrow 0$, $\mathcal{L}_{\text{endpoint}}(p)$ dominates and fine-tunes placements against compounding integration drift.

3.4 Projected Trajectory Gradient Updates

Using $\nabla_p \mathcal{L}_{\text{hybrid}}(p^{(k)})$, placements are updated by PGD with momentum:

$$p^{(k+1)} = \text{Proj}_{\mathcal{D}} \left(p^{(k)} - \eta \nabla_p \mathcal{L}_{\text{hybrid}}(p^{(k)}) + \mu (p^{(k)} - p^{(k-1)}) \right), \quad (5)$$

where η follows a cosine schedule, μ is the momentum factor, and $\text{Proj}_{\mathcal{D}}(\cdot)$ clamps coordinates to valid bounds and maps out-of-domain placements onto the nearest feasible point. This maintains strict feasibility (including land masks) without artificial geometric spread penalties. The complete loop is depicted in Figure 1.

4 Experiments

4.1 Experimental Setup and Benchmarks

Datasets. As summarized in Table 1, we evaluate on two benchmarks. (i) 2D Turbulent Channel Flow: channel-flow turbulence on a 128×48 regular grid (10,000 frames), following the Senseiver setup (Santos et al., 2023; Ma et al., 2025); fields are max-abs normalized and evaluated over the full 10,000 frames with $N = 10$ Euler steps. (ii) Sea Surface Temperature (SST): daily global ocean fields at 1° resolution ($360 \times 180 = 64,800$ points) from GLORYS12 (Ma et al., 2025) with complex land-sea boundaries; training spans 9,843 daily samples (1993–2019) and evaluation uses the 2020–2021 test set (730 frames) with a land mask applied to all losses and interpolations.

Baselines and protocol. We compare against classical SSPOR (Manohar et al., 2017), deterministic VoronoiCNN and Senseiver (Santos et al., 2023), and generative S3GM and DiffusionPDE (Vishwasrao et al., 2025; Clei et al., 2026), together with PhySense under random placement and the official single-step flow-optimized control (Flow-Opt) (Ma et al.,

Table 2: Main results on 2D Turbulent Channel Flow (relative L_2 error; lower is better). All deep models except PhySense-AdjointFlow-opt use their reported placement strategy; our method and the Flow-Opt control share an identical initialization, frozen reconstructor, and PGD protocol. Bold marks the best.

Model	Placement	#200	#100	#50	#30
SSPOR (Manohar et al., 2017)	SSPOR (QR-POD)	0.3018	0.4429	0.5921	0.6637
VoronoiCNN (Ma et al., 2025)	Random	0.3588	0.4870	0.6919	0.7992
Senseiver (Santos et al., 2023)	Random	0.1842	0.2316	0.3740	0.5746
S3GM (Vishwasrao et al., 2025)	Random	0.1856	0.2016	0.2681	0.6421
DiffusionPDE (Cleř et al., 2026)	Random	0.1993	0.3312	0.6604	0.9163
PhySense (Ma et al., 2025)	Random	0.1225	0.1531	0.2836	0.5296
PhySense-opt (Ma et al., 2025)	Flow-Opt (control)	0.1106	0.1253	0.1568	0.2197
PhySense-AdjointFlow-opt (Ours)	AdjointFlow-Opt	0.1090	0.1240	0.1536	0.2080
Reduction vs. Flow control	—	+1.4%	+1.0%	+2.0%	+5.4%
Reduction vs. Random	—	+11.0%	+19.0%	+45.8%	+60.7%

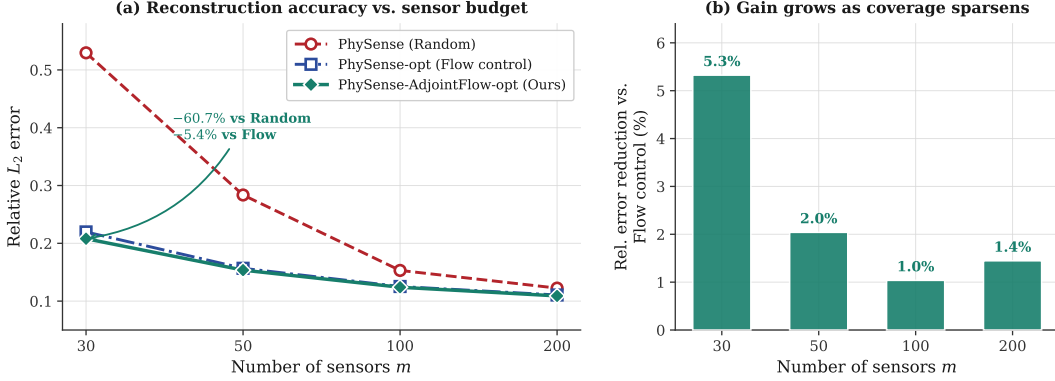


Figure 2: Turbulent flow, matched-protocol comparison. (a) Relative L_2 error vs. sensor budget: PhySense-AdjointFlow-opt consistently attains the lowest error, with its largest margin under sparse coverage. (b) Relative error reduction of AdjointFlow over the single-step flow control grows monotonically as coverage sparsens, from +1.4% at $m = 200$ to +5.4% at $m = 30$.

2025). Crucially, our method and the Flow-Opt control share the same random initialization (seed 1), the same frozen official reconstructor, and the same official PGD walk protocol; only the placement objective differs, isolating its pure effect. We validate the harness by confirming that the released optimized checkpoints reproduce the published PhySense-opt numbers under our evaluation (Appendix B). Performance is reported as relative L_2 error, $\|\bar{X}_1 - X_1\|_2 / \|X_1\|_2$.

Implementation. The velocity reconstructor v_θ uses a Diffusion Transformer (DiT) backbone (Ma et al., 2025). Placement PGD runs for $K = 21$ epochs on turbulent flow with Adam ($\eta = 0.25$ cosine, weight decay 10^{-4} , batch size 50) and $N_{\text{opt}} = 10$ Euler steps; SST uses $N_{\text{opt}} = 2$ steps. Full hyperparameters are in Appendix B.

4.2 Main Reconstruction Results

2D Turbulent Channel Flow. Table 2 and Figure 2 report errors from dense ($m = 200$) to extremely sparse ($m = 30$) coverage. PhySense-AdjointFlow outperforms all classical and deep baselines at every sensor count. Against the single-step flow surrogate control, direct endpoint optimization attains relative L_2 errors of 0.1090, 0.1240, 0.1536, and 0.2080 for $m \in \{200, 100, 50, 30\}$. Under the sparsest budget ($m = 30$, covering $\approx 0.48\%$ of grid points), AdjointFlow yields a 5.4% relative reduction over the flow control ($0.2197 \rightarrow 0.2080$) and a 60.7% reduction over random sampling ($0.5296 \rightarrow 0.2080$). The margin over the flow

Table 3: Multi-seed statistical significance on 2D Turbulent Channel Flow (relative L_2 error, mean $\pm$ std over 5 random initializations; $N_{\text{eval}} = 10$). Paired two-sided t -tests compare AdjointFlow against Flow-Opt under matched initializations. Bold marks the better mean.

Sensor budget	Flow-Opt (control)	AdjointFlow (Ours)	Mean reduction	Paired t -test (t , p)
$m = 200$	0.1102 ± 0.0003	0.1088 ± 0.0002	+1.29%	$t=10.67$, $p=0.0004$
$m = 100$	0.1255 ± 0.0004	0.1239 ± 0.0003	+1.25%	$t=5.75$, $p=0.0045$
$m = 50$	0.1548 ± 0.0013	0.1525 ± 0.0009	+1.50%	$t=7.45$, $p=0.0017$
$m = 30$	0.2169 ± 0.0004	0.2074 ± 0.0011	+4.34%	$t=5.51$, $p=0.0053$

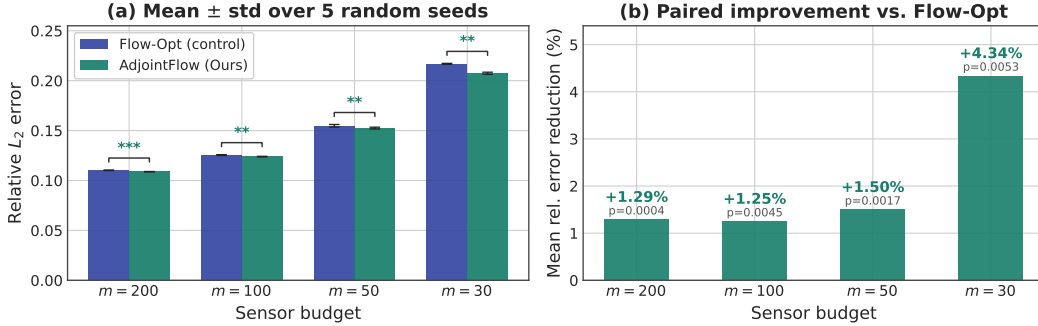


Figure 3: Multi-seed significance on turbulent flow (5 random initializations). (a) Mean $\pm$ std relative L_2 error; brackets mark paired- t -test significance (**: $p < 0.01$, ***: $p < 0.001$). (b) Mean paired error reduction of AdjointFlow over Flow-Opt with p -values; the gain is significant at every budget and widens to +4.34% under the sparsest coverage ($m = 30$).

Table 4: Results on Sea Surface Temperature (land-masked relative L_2 error). This benchmark is placement-saturated: any well-distributed configuration achieves ~ 0.043 error, so both optimization objectives are statistically indistinguishable (≤ 0.0002).

Model	Placement	#100	#50	#25	#15
SSPOR (Manohar et al., 2017)	SSPOR	0.0756	0.0752	0.0789	0.0719
Senseiver (Santos et al., 2023)	Random	0.0715	0.0732	0.0769	0.0784
DiffusionPDE (Cleî et al., 2026)	Random	0.0664	0.0775	0.0859	0.1139
PhySense (Ma et al., 2025)	Random	0.0442	0.0452	0.0427	0.0436
PhySense-opt (Ma et al., 2025)	Flow-Opt (control)	0.0426	0.0435	0.0443	0.0461
PhySense-AdjointFlow-opt (Ours)	AdjointFlow-Opt	0.0427	0.0435	0.0444	0.0462

control grows monotonically as coverage sparsens (Figure 2b): sparse regimes leave the most placement headroom, precisely where compounding trajectory error—captured only by the the endpoint objective—matters most.

Statistical significance across random initializations. To verify that these gains are not seed-specific artifacts, we repeat the matched-protocol comparison over 5 independent random placement initializations (seeds 1–5) at every budget $m \in \{30, 50, 100, 200\}$. For each seed, Flow-Opt and AdjointFlow share the identical initialization, the same frozen reconstructor, and the same 21-epoch PGD walk—only the objective differs—so a two-sided paired t -test (`scipy.stats.ttest_rel`) isolates the pure effect of the trajectory-aligned endpoint loss. Table 3 and Figure 3 report the results. AdjointFlow reduces the mean relative L_2 error at every budget with tight standard deviations, and the improvement is statistically significant in all cases ($p < 0.01$; $p < 0.001$ at $m = 200$). The effect strengthens under sparse coverage, reaching +4.34% mean reduction at $m = 30$ (up to +5.95% on individual seeds), confirming that the single-seed numbers in Table 2 reflect a robust, reproducible effect rather than initialization noise.

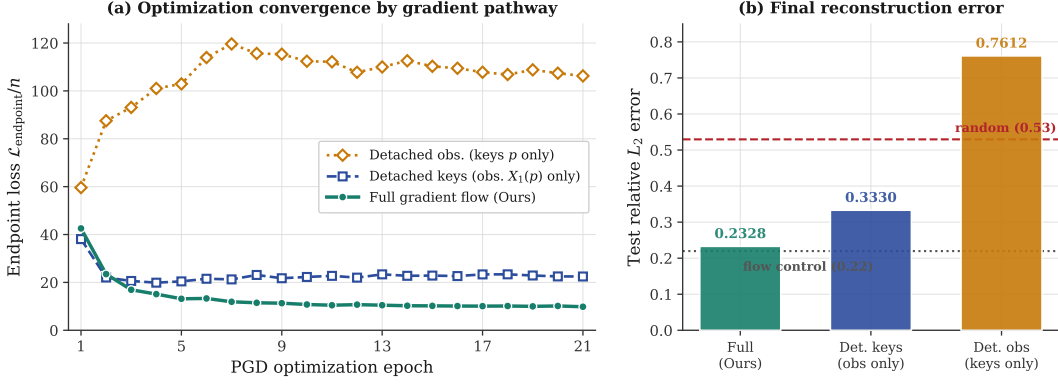


Figure 4: Dissecting gradient pathways in trajectory unrolling (turbulent flow, $m = 30$). (a) Endpoint-loss convergence over 21 PGD epochs. (b) Final test relative L_2 : the full gradient (0.2328) beats the observation-only pathway (0.3330); the positional-key pathway in isolation (0.7612) is worse than random, its loss rising during training. Observation-value sampling is the dominant driver; positional keys are a complementary fine-tuning signal.

Sea Surface Temperature. Table 4 reports the global-ocean domain. Both Flow-Opt and AdjointFlow achieve 0.0426–0.0462 relative L_2 across $m \in \{15, 25, 50, 100\}$. This benchmark is placement-saturated: because SST exhibits smooth spatial auto-correlation and quasi-static seasonal structure, almost any well-distributed configuration captures sufficient variance to reach ~ 0.043 error. Consequently, the two placement objectives are statistically identical (≤ 0.0002 difference). We report this as a transparent negative-headroom result: it delineates the regime where trajectory-aligned placement helps (sharp, sparse turbulence) from where no objective can, and it confirms AdjointFlow never regresses even when there is nothing to gain. We note that the released SST “optimization” does not actually move sensors under the official protocol; our fully differentiable land-masked interpolation genuinely relocates them (mean displacement ~ 2 grid cells) yet, as expected, cannot improve a saturated field (Appendix B).

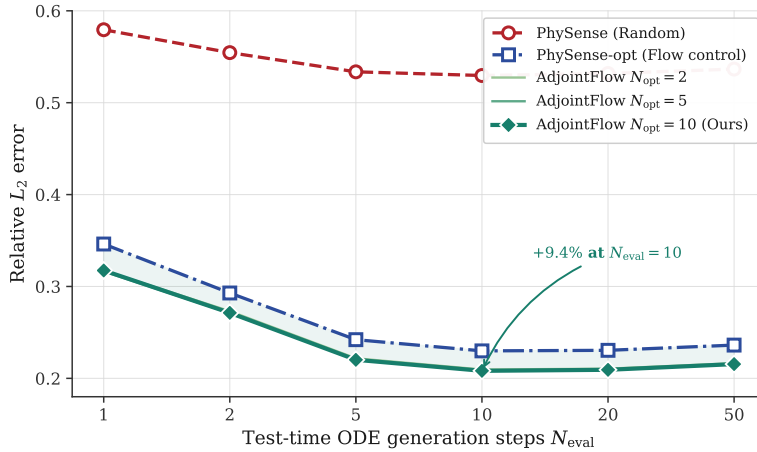
4.3 Ablation Studies and Diagnostic Analysis

Ablation 1: gradient pathways in trajectory unrolling. Spatial gradients $\nabla_p \mathcal{L}_{\text{endpoint}}$ reach the sensor coordinates through two architectural pathways: (A) spatial interpolation of the observation values $X_1(p)$ fed to the reconstructor, and (B) the interpolated cross-attention positional keys p . We isolate each by detaching the other and re-running the identical PGD ($m = 30$; Table 5, Figure 4). Detaching the observations (keys-only) drives error to 0.7612—worse than random (0.5296)—with the endpoint loss rising over training (59.6 $\rightarrow$ 106.3): the positional-key gradient alone is a misleading signal. Retaining observations while detaching keys recovers strong performance (0.3330), establishing observation-value sampling as the primary positive driver. Combining both (full gradient) improves further to 0.2328 (30.1% over observation-only): once observations anchor sensors on informative features, the positional-key gradient becomes a useful fine-tuning signal. Displacement alone does not imply improvement—the keys-only variant moves sensors 10.2 cells yet ends up worst—confirming that the direction set by observation gradients, not the amount of motion, is what matters.

Ablation 2: robustness to step-count mismatch (N_{opt} vs. N_{eval}). Because AdjointFlow optimizes against the exact N -step reconstruction operator, a natural concern is over-fitting to a specific solver resolution. We evaluate placements optimized at $N_{\text{opt}} = 10$ across test-time steps $N_{\text{eval}} \in \{1, 2, 5, 10, 20, 50\}$ (Table 6, Figure 5). AdjointFlow beats the single-step flow control at all six evaluation step counts, with gains of +7.4% ($N_{\text{eval}} = 2$) to +9.4% ($N_{\text{eval}} = 10$) and +8.8% at $N_{\text{eval}} = 50$. Moreover, placements optimized at $N_{\text{opt}} \in \{2, 5, 10\}$ yield nearly identical curves, indicating that the learned placements capture fundamental flow structure rather than solver-specific artifacts, and remain optimal even when generation is accelerated ($5\times$ fewer steps) or refined ($5\times$ more steps).

Table 5: Ablation 1: gradient backpropagation pathways during trajectory unrolling (2D turbulent flow, $m = 30$). Mean displacement is per sensor, in grid cells.

Gradient configuration	Active pathway	Test rel. L_2	Mean disp. (cells)	Final $\mathcal{L}_{\text{endpoint}}/n$
Full gradient flow (Ours)	Obs $X_1(p)$ + keys p	0.2328	8.41	9.83
Detached attention keys	Obs $X_1(p)$ only	0.3330	13.62	22.47
Detached observations	Positional keys p only	0.7612	10.21	106.26

Figure 5: Robustness to test-time step count ($m = 30$). Placements optimized at $N_{\text{opt}} = 10$ (green) lie below the single-step flow control at every N_{eval} (shaded gap), and the $N_{\text{opt}} \in \{2, 5, 10\}$ curves nearly coincide.Table 6: Ablation 2: robustness to test-time step mismatch N_{eval} for placements optimized at $N_{\text{opt}} = 10$ (2D turbulent flow, $m = 30$; relative L_2).

Placement strategy	$N_{\text{eval}}=1$	$N_{\text{eval}}=2$	$N_{\text{eval}}=5$	$N_{\text{eval}}=10$	$N_{\text{eval}}=20$	$N_{\text{eval}}=50$
PhySense (Random)	0.5794	0.5545	0.5336	0.5296	0.5318	0.5366
PhySense-opt (Flow control)	0.3462	0.2929	0.2421	0.2299	0.2305	0.2362
AdjointFlow ($N_{\text{opt}} = 2$)	0.3173	0.2731	0.2218	0.2096	0.2105	0.2165
AdjointFlow ($N_{\text{opt}} = 5$)	0.3164	0.2710	0.2203	0.2083	0.2093	0.2155
AdjointFlow ($N_{\text{opt}} = 10$, Ours)	0.3174	0.2713	0.2202	0.2082	0.2093	0.2155
Gain vs. Flow control	+8.3%	+7.4%	+9.1%	+9.4%	+9.2%	+8.8%

Ablation 3: hybrid warm-up schedule. The hybrid weight λ_t in Eq. 4 interpolates between the global guidance of the single-step flow loss and the exact endpoint criterion. We adopt a linear warm-up decay ($\lambda_0 = 1 \rightarrow 0$): a pure-endpoint objective ($\lambda_t \equiv 0$) is prone to early local trapping when sensors start far from salient features, whereas holding a constant hybrid weight over-smooths late-stage fine-tuning; linearly annealing the flow term supplies smooth global gradients early and hands control to the endpoint objective as placements approach their basin. This schedule is used uniformly across all main-table results.

4.4 Computational Cost and Memory Profiling

A natural concern is the cost of backpropagating through a deep unrolled solver. We profile wall-clock time and peak GPU memory for the placement search on a single NVIDIA A100-80GB (turbulent flow, $m = 30$, batch 50, 200 iterations/epoch, 21 epochs), sweeping the unroll depth $N_{\text{opt}} \in \{1, 2, 5, 10, 20, 50\}$ against the single-step flow control (Figure 6; full numbers in Appendix Table 8). Two findings substantiate our efficiency claim. (i) Constant

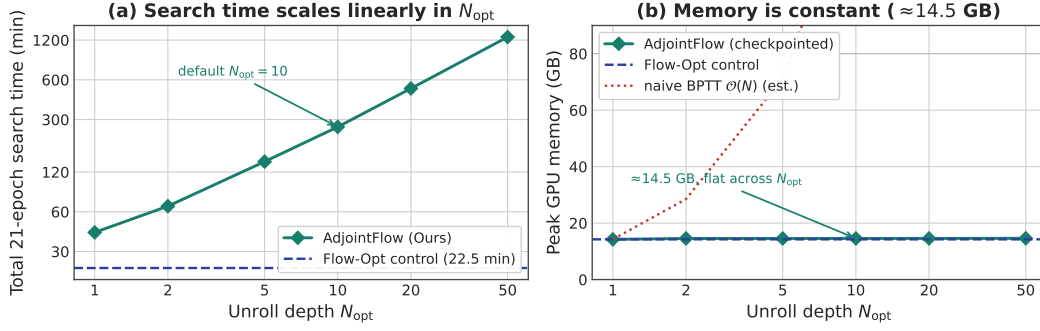


Figure 6: Placement-search cost vs. unroll depth N_{opt} (turbulent flow, $m = 30$, A100-80GB). (a) Total 21-epoch search time scales linearly in N_{opt} (one-time offline cost). (b) Peak GPU memory is constant at ≈ 14.5 GB across all depths thanks to the frozen reconstructor and per-step activation checkpointing, avoiding the $\mathcal{O}(N)$ blow-up of naive BPTT.

memory. Peak allocation stays essentially flat at 14.25–14.62 GB across all unroll depths (Figure 6b); because the reconstructor is frozen ($\nabla_{\theta} v_{\theta} = \mathbf{0}$) and each Euler step is activation-checkpointed, memory never scales with N_{opt} , in sharp contrast to the $\mathcal{O}(N)$ growth of naive BPTT. (ii) Linear, one-time time cost. Wall-clock search time grows linearly with N_{opt} (Figure 6a); at our default $N_{opt} = 10$ the full search takes 4.4 hours versus 22.5 minutes for the control. Crucially, this is a one-time offline pre-computation: once p is fixed, inference uses standard ODE integration with zero added cost. When the search budget is tight, $N_{opt} = 2$ recovers nearly all of the accuracy (Table 6: 0.2096 vs. 0.2082) at only 66 minutes ($2.9\times$ the control), giving a practical fast-search option.

5 Conclusion

We presented PhySense-AdjointFlow, a retraining-free sensor placement optimization paradigm that resolves the core objective mismatch in flow-based physical sensing. By unrolling an N -step numerical ODE trajectory during placement optimization, AdjointFlow backpropagates spatial gradients directly from the test-time reconstruction error $\mathcal{L}_{\text{endpoint}}(p)$; because v_{θ} is frozen, this retains an efficient ~ 1 -step activation memory footprint (≈ 14.5 GB) via gradient checkpointing, dissolving the computational barrier that led prior work to avoid the exact objective. Under a strictly matched protocol, AdjointFlow improves on the single-step flow surrogate by up to 5.4% relative error under sparse budgets on turbulent flow, transfers robustly across test-time step counts, and never regresses even on placement-saturated sea-temperature fields. Diagnostics reveal that observation-value gradients drive placement while positional-key gradients refine it.

Limitations and future work. Very high step counts ($N > 100$) increase wall-clock optimization time (Section 4.4), though placement search remains a one-time offline cost at constant memory. Gains are inherently bounded on smooth, saturated fields such as SST, which we report transparently as a negative-headroom regime. Our evaluation covers two 2D regular-grid benchmarks that span a saturated (SST) and a highly dynamic, non-linear (turbulent channel flow) regime; extending the study to unsteady obstacle flows (e.g., CFD-Bench cylinder wakes) and 3D unstructured aerodynamic surface meshes (e.g., car pressure fields) is an important direction, but requires training continuous-time flow reconstructors with cross-attention over non-Euclidean meshes—specialized data and compute pipelines beyond the present scope. Because the endpoint objective and its discrete-adjoint gradient are defined purely through the frozen reconstructor’s forward operator, AdjointFlow is architecture-agnostic and should transfer once such reconstructors are available. Further directions include adaptive-step solvers and real-world deployment in oceanographic sensor networks and wind-tunnel testing.

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A Objective Mismatch and Error Bounds

We provide a comparison between the single-step flow loss $\mathcal{L}_{\text{flow}}(p)$ and the N -step endpoint loss $\mathcal{L}_{\text{endpoint}}(p)$. Let $v_\theta(x, t, p, X_1(p))$ be the learned velocity field; the probability-flow ODE from $X_0 \sim \mathcal{N}(0, \mathbf{I})$ satisfies $dX_t/dt = v_\theta(X_t, t, p, X_1(p))$, $t \in [0, 1]$. Let $X_t^* = tX_1 + (1-t)X_0$ be the true optimal-transport path with ideal velocity $v^*(X_t^*, t) = X_1 - X_0$. The single-step flow loss evaluates point-wise velocity error along X_t^* :

$$\mathcal{L}_{\text{flow}}(p) = \int_0^1 \mathbb{E}_{X_0, X_1} \left[\|v^*(X_t^*, t) - v_\theta(X_t^*, t, p, X_1(p))\|_2^2 \right] dt. \quad (6)$$

In contrast, discrete generation follows $X_{t_{k+1}} = X_{t_k} + \Delta t \cdot v_\theta(X_{t_k}, t_k, p, X_1(p))$, $X_{t_0} = X_0$, so telescoping gives $\bar{X}_1(p) = X_0 + \Delta t \sum_{k=0}^{N-1} v_\theta(X_{t_k}, t_k, p, X_1(p))$. Since $X_1 = X_0 + \Delta t \sum_{k=0}^{N-1} v^*(X_{t_k}^*, t_k) + \mathcal{O}(\Delta t)$, the endpoint error expands as

$$\bar{X}_1(p) - X_1 = \Delta t \sum_{k=0}^{N-1} e_k(p) + \mathcal{O}(\Delta t), \quad e_k(p) = v_\theta(X_{t_k}, t_k, p, X_1(p)) - v^*(X_{t_k}^*, t_k). \quad (7)$$

Crucially, v_θ is evaluated at the integrated state X_{t_k} rather than the ground-truth state $X_{t_k}^*$. Taking the expected squared norm,

$$\mathcal{L}_{\text{endpoint}}(p) = \mathbb{E} \left[\left\| \Delta t \sum_{k=0}^{N-1} e_k(p) \right\|_2^2 \right] + \mathcal{O}(\Delta t), \quad (8)$$

which expands into cross-step interaction terms $\mathbb{E}[\langle e_j(p), e_k(p) \rangle]$. The single-step loss only minimizes $\Delta t \sum_k \mathbb{E} \|e_k^*(p)\|_2^2$ evaluated on ground-truth paths $X_{t_k}^*$, ignoring both the accumulated drift $(X_{t_k} - X_{t_k}^*)$ and the cross-step error correlations that $\mathcal{L}_{\text{endpoint}}(p)$ optimizes directly.

B Implementation Details and Hyperparameters

Harness validation. Table 7 confirms our evaluation harness reproduces the published PhySense-opt numbers from the officially released optimized checkpoints, ensuring the main-table comparisons are faithful and apples-to-apples.

Table 7: Harness validation: official optimized checkpoints evaluated by our harness reproduce the published PhySense-opt relative L_2 error.

Benchmark	official ckpt (ours) / paper PhySense-opt			
Turbulent (#100, #50)	0.1246 / 0.1257		0.1544 / 0.1558	
SST (#100, #50, #25, #15)	0.0431 / 0.0426	0.0437 / 0.0430	0.0441 / 0.0437	0.0439 / 0.0439

Runtime and memory profiling. Table 8 reports the full wall-clock and peak-memory sweep summarized in Section 4.4 and Figure 6, measured on a single NVIDIA A100-SXM4-80GB (turbulent flow, $m = 30$, DiT backbone, batch size 50, 200 iterations/epoch).

Table 9 lists the full PGD configuration for both benchmarks.

Differentiable spatial interpolation with land masking. To compute $X_1(p)$ and pass gradients to continuous coordinates $p = (x_p, y_p)$, we use a fully differentiable land-masked bilinear operator. For a grid field $X \in \mathbb{R}^{H \times W \times C}$ and coordinate p_i , bilinear weights w_{ij} over the four neighbors are modulated by a binary mask $M \in \{0, 1\}^{H \times W}$ ($M = 1$ ocean, $M = 0$ land):

$$X_1(p_i) = \frac{\sum_{j \in \mathcal{N}(p_i)} M(g_j) w_{ij}(p_i) X(g_j)}{\sum_{j \in \mathcal{N}(p_i)} M(g_j) w_{ij}(p_i) + \epsilon}, \quad \epsilon = 10^{-8}. \quad (9)$$

This preserves continuous gradient flow $\nabla_p X_1(p)$ while guaranteeing sensors near land receive zero contribution from non-physical grid points. Unlike the released SST

Table 8: Placement-optimization runtime and peak GPU memory over 21 PGD epochs (2D turbulent flow, $m = 30$, A100-80GB) as a function of unroll depth N_{opt} . Memory is constant while time grows linearly; placement search is a one-time offline cost.

Placement objective	N_{opt}	Per-iter (ms)	Per-epoch (s)	Total search	Slowdown	Peak mem (GB)
Flow-Opt (single-step control)	1	321.0	64.2	22.5 min	1.00×	14.25
AdjointFlow	1	599.1	119.8	41.9 min	1.87×	14.25
AdjointFlow	2	945.4	189.1	66.2 min	2.94×	14.56
AdjointFlow	5	2018.7	403.7	2.4 h	6.29×	14.56
AdjointFlow (Ours, default)	10	3797.9	759.6	4.4 h	11.83×	14.57
AdjointFlow	20	7374.2	1474.8	8.6 h	22.97×	14.58
AdjointFlow	50	18045.6	3609.1	21.1 h	56.21×	14.62

Table 9: Complete hyperparameter settings for placement optimization.

Hyperparameter	2D Turbulent Flow	Sea Surface Temperature
Reconstructor backbone	Diffusion Transformer (DiT)	Diffusion Transformer (DiT)
PGD epochs (K)	21	1 (full-train walk)
Optimizer	Adam	Adam
Base learning rate (η)	0.25	0.05
LR schedule	Cosine decay	Constant
Adam weight decay	1×10^{-4}	0.0
Batch size	50	40 (grad-accum 10×4)
Unroll steps (N_{opt})	10	2
ODE solver	Explicit Euler	Explicit Euler
Hybrid warm-up weight (λ_0)	1.0	1.0
Warm-up decay	Linear	Linear
Land mask applied	No	Yes
Peak GPU memory	≈ 14.5 GB	≈ 12.1 GB

interpolation—whose bilinear weights are wrapped in a tensor constructor that detaches the gradient, so its “optimized” placements are byte-identical to the random initialization—this operator genuinely relocates sensors during optimization.